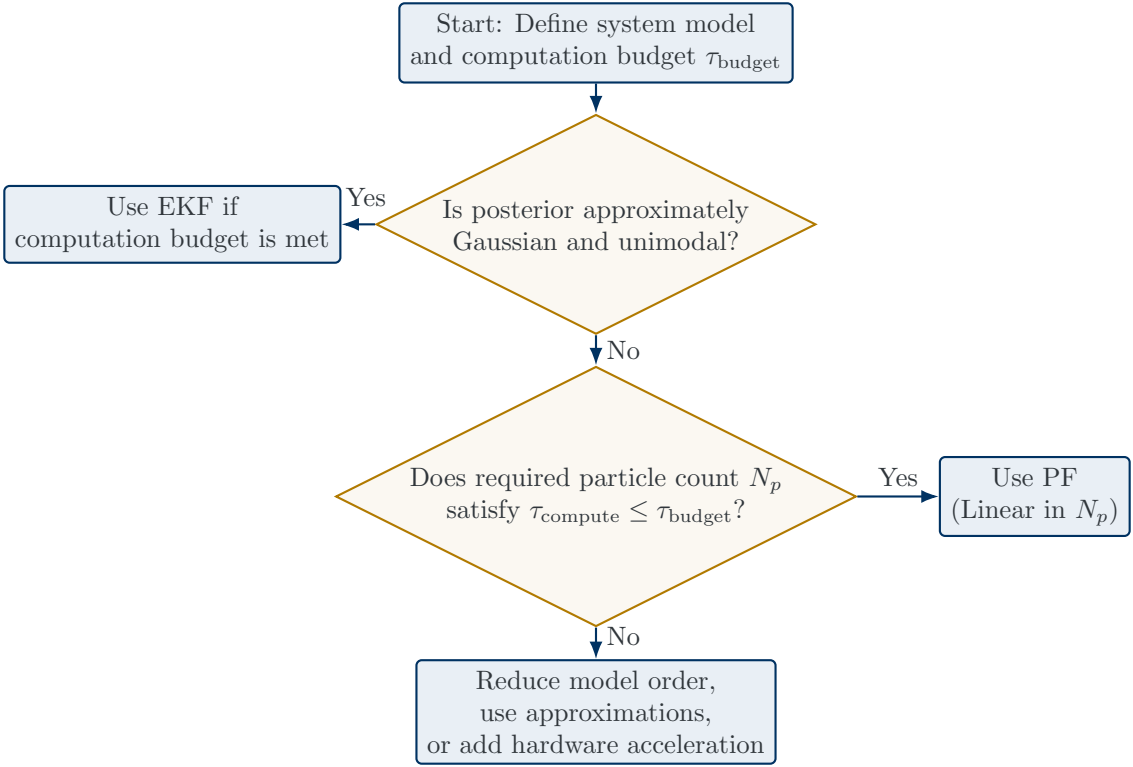
Start: Define system model and computation budget $\tau_{\text{budget}}$
Is posterior approximately Gaussian and unimodal?
Yes
Use EKF if computation budget is met
No
Does required particle count $N_p$ satisfy $\tau_{\text{compute}} \leq \tau_{\text{budget}}$?
Yes
Use PF (Linear in $N_p$)
No
Reduce model order, use approximations, or add hardware acceleration